# Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

## Section 1: Domains and Ranges of Inverse Trig Functions

**1** What are the domain and range of the principal arcsine function $y = \arcsin x$?

**a.** Domain: $[-1, 1]$; Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

**b.** Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

**c.** Domain: $[-1, 1]$; Range: $[0, \pi]$

**d.** Domain: $(-\infty, \infty)$; Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

**2** What are the domain and range of $y = \arccos x$?

**a.** Domain: $[-1, 1]$; Range: $[0, \pi]$

**b.** Domain: $[-1, 1]$; Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

**c.** Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

**d.** Domain: $[0, 1]$; Range: $[0, \pi/2]$

**3** What are the horizontal asymptotes of the graph of $y = \arctan x$?

**a.** $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$

**b.** $y = \pi$ and $y = 0$

**c.** $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$

**d.** $y = 1$ and $y = -1$

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## Section 2: Derivatives of Inverse Trigonometric Functions

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**4** What is the derivative of $f(x) = \arcsin(3x)$ for $|x| < \frac{1}{3}$?

**a.** $\frac{3}{\sqrt{1-9x^2}}$

**b.** $\frac{1}{\sqrt{1-9x^2}}$

**c.** $\frac{3}{\sqrt{1-3x^2}}$

**d.** $\frac{3}{1+9x^2}$

**5** Find the derivative of $g(x) = \arctan\left(\frac{x}{2}\right)$.

**a.** $\frac{2}{4+x^2}$

**b.** $\frac{1}{1+x^2/4}$

**c.** $\frac{1}{4+x^2}$

**d.** $\frac{2}{\sqrt{4-x^2}}$

**6** What is the relationship between the derivatives of $\arcsin x$ and $\arccos x$?

**a.** $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x] = -\frac{1}{\sqrt{1-x^2}}$

**b.** $\frac{d}{dx}[\arccos x] = \frac{d}{dx}[\arcsin x]$

**c.** $\frac{d}{dx}[\arccos x] = \frac{1}{1-x^2}$

**d.** They are reciprocals of each other

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## Section 3: Tangent Lines and Extreme Values

**7** Find the slope of the tangent line to $y = \arcsin x$ at $x = 0$.

**a.** 1

**b.** 0

**c.** $\pi/2$

**d.** Undefined

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**8** What happens to the slope of $y = \arcsin x$ as $x \to 1^-$?

**a.** The slope approaches $+\infty$ (vertical tangent)

**b.** The slope approaches $0$

**c.** The slope approaches $1$

**d.** The slope approaches $-1$

**9** Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

**a.** $y - \frac{\pi}{4} = \frac{1}{2}(x - 1)$

**b.** $y - \frac{\pi}{4} = 1(x - 1)$

**c.** $y = \frac{1}{2}x + \frac{\pi}{4}$

**d.** $y - 1 = \frac{1}{2}(x - \frac{\pi}{4})$

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## Section 4: Integration Producing Inverse Trig Functions

**10** Evaluate $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}dx$.

**a.** $\frac{\pi}{6}$

**b.** $\frac{\pi}{3}$

**c.** $\frac{\pi}{4}$

**d.** $\frac{1}{2}$

**11** Evaluate $\int_0^1 \frac{1}{1+x^2}dx$.

**a.** $\frac{\pi}{4}$

**b.** $\frac{\pi}{2}$

**c.** $1$

**d.** $\ln 2$

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**12** Evaluate $\int \frac{1}{\sqrt{4-9x^2}} dx$.

**a.** $\frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + C$

**b.** $\frac{1}{2}\arcsin\left(\frac{3x}{2}\right) + C$

**c.** $\arcsin\left(\frac{3x}{2}\right) + C$

**d.** $\frac{1}{6}\arcsin\left(\frac{3x}{2}\right) + C$

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